



EXAM PAPERS PRACTICE

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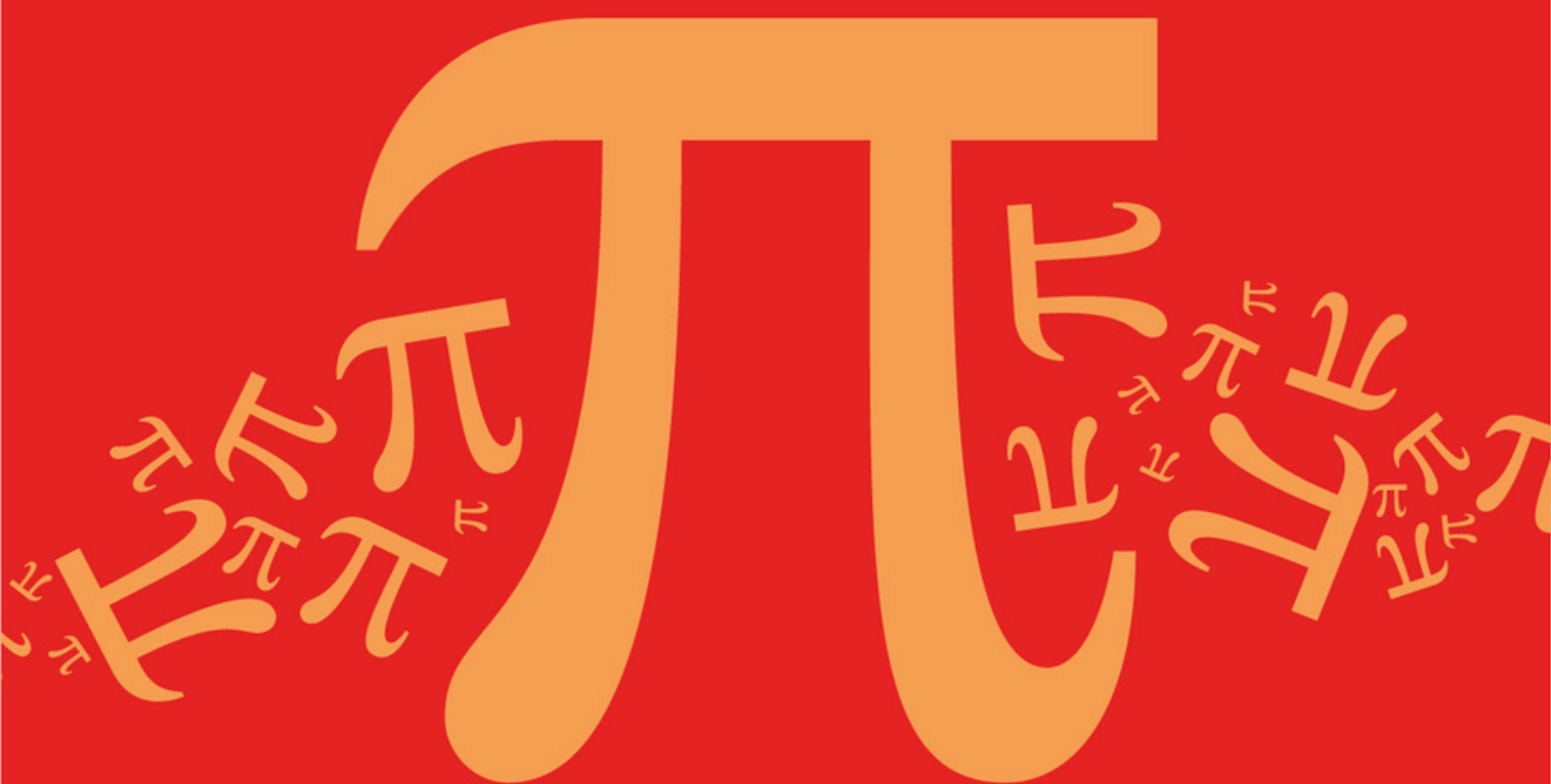
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C2: Zero and Identity

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

C: Matrices

Sub topic

C2: Zero and Identity

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses factorisation or pre-multiplication to isolate A	AO3.1a	M1	$\mathbf{AB} - 2\mathbf{A} = \mathbf{I}$ $\mathbf{A}(\mathbf{B} - 2\mathbf{I}) = \mathbf{I}$
Deduces A in terms of B and I . Could be implied by sight of $\begin{bmatrix} 1 & -2 \\ -4 & 6 \end{bmatrix}$ with attempt to invert.	AO2.2a	A1	$\mathbf{A} = (\mathbf{B} - 2\mathbf{I})^{-1}$
Obtains correct matrix A .	AO1.1b	A1	$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ -4 & 6 \end{bmatrix}^{-1}$ $\mathbf{A} = \frac{1}{-2} \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} -3 & -1 \\ -2 & -0.5 \end{bmatrix}$

ALT

Marking Instructions	AO	Marks	Typical Solution
Sets up four equations with at least three correct.	AO3.1a	M1	Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 2 \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $3a - 4b = 1 + 2a$ and $3c - 4d = 0 + 2c$ and $-2a + 8b = 0 + 2b$ and $-2c + 8d = 1 + 2d$
Deduces at least two correct elements of A . Note: Two correct elements from just two correct equations can score M1A1.	AO2.2a	A1	$a = 4b + 1$ and $c = 4d$ $6b = 2a$ and $6d = 2c + 1$ $\therefore 3b = 4b + 1$ and $6d = 2(4d) + 1$ $-1 = b$ and $-1 = 2d \Rightarrow d = -\frac{1}{2}$ $\therefore a = 4 \times -1 + 1$ and $c = 4 \times -\frac{1}{2}$ $a = -3$ and $c = -2$
Obtains correct matrix A	AO1.1b	A1	$\therefore \mathbf{A} = \begin{bmatrix} -3 & -1 \\ -2 & -0.5 \end{bmatrix}$
Total 3 marks			

Q2.

(a) (i) $X^2 = \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix}; \quad (m \Rightarrow) 7$
(m $\Rightarrow$) 7 or 7 as top left element of X^2

B1

1

(ii) $X^3 = \begin{bmatrix} 13 & 14 \\ 21 & 6 \end{bmatrix};$
At least 2 elements correct

M1

$7X = \begin{bmatrix} 7 & 14 \\ 21 & 0 \end{bmatrix}$
PI

B1

$X^3 - 7X = \begin{bmatrix} 13-7 & 14-14 \\ 21-21 & 6-0 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$
Ft on c's m value

A1F

$= 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 6I$

CSO Accept either form but at least one must be shown explicitly

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A1

4

(b) (i) Reflection in the x-axis
OE

B1

1

(ii) $B = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
Either OE. For M mark, accept dec. equiv.
(at least 3sf) for $\frac{1}{\sqrt{2}}$

M1

$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

NMS SC1 for $k = \frac{1}{\sqrt{2}}$ or better.

A1

2

(iii) $\mathbf{AB} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = k \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Attempt to find $\mathbf{AB} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

M1

$$= k \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \left\{ \text{or } k \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$$

Either $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

A1

$$= k \begin{bmatrix} -3 \\ -1 \end{bmatrix}$$

Completing the matrix mult. to reach a 2×1 matrix

m1

(Image of P is the point) $\left(-\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$

CSO

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SC Wrong order, works with $\mathbf{BA} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$,

mark out of a max of M1A0 m1A0

A1

4

[12]

Q3.

(a) $\mathbf{A}^2 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

$$= \begin{bmatrix} -1 & -2 & -4 \\ 5 & 6 & 4 \\ 10 & 10 & 9 \end{bmatrix}$$

$$p = -1$$

p-value

B1

$$q = 10$$

q-value

B1

2

- (b) $\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = \mathbf{0}$ multiply by $\mathbf{A}^{-1}$
 $(\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I})\mathbf{A}^{-1} = (\mathbf{0})\mathbf{A}^{-1}$
Multiplication by $\mathbf{A}^{-1}$

M1

$$\mathbf{A}^3\mathbf{A}^{-1} - 6\mathbf{A}^2\mathbf{A}^{-1} + 11\mathbf{A}\mathbf{A}^{-1} - 6\mathbf{I}\mathbf{A}^{-1} = \mathbf{0}$$

$$\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I} - 6\mathbf{A}^{-1} = \mathbf{0}$$

$$6\mathbf{A}^{-1} = \mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I}$$

$$\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I})$$

AG

A1

2

(c) $\mathbf{A}^{-1} = \frac{1}{6} \begin{bmatrix} 4 & -2 & 2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix}$

$$r = 4$$

r-value

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B1

$$s = -2$$

s-value

B1

2

(d) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4 & -2 & 2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix} \begin{bmatrix} k \\ 5 \\ 7 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4k - 10 + 14 \\ -k + 25 - 14 \\ -2k - 10 + 14 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4k + 4 \\ 11 - k \\ 4 - 2k \end{bmatrix}$

use of $\mathbf{A}^{-1} \mathbf{v}$ – one row correct

M1

$$x = \frac{2k+2}{3}, y = \frac{11-k}{6}, z = \frac{2-k}{3}$$

correct solution for one variable

A1

all correct CAO

A1

Alternative:

If solving equations by elimination, M1 A1 for correct solution for one variable, A1 all correct

3

[9]

Q4.

(a) (i) $\mathbf{A} - \mathbf{B} = \begin{bmatrix} p-3 & 1 \\ 2 & p-3 \end{bmatrix}$

B1

1

(ii) $\mathbf{AB} = \begin{bmatrix} p & 2 \\ 4 & p \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3p+4 & p+6 \\ 12+2p & 4+3p \end{bmatrix}$

Finding $\mathbf{AB}$ and at least 2 elements correct

M1

CSO

A1

2

(b) $\mathbf{A} - \mathbf{B} + \mathbf{AB} = \begin{bmatrix} 4p+1 & p+7 \\ 14+2p & 1+4p \end{bmatrix}$

Only ft if all matrices are 2 by 2

PI by later correct work

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B1F

$\mathbf{A} - \mathbf{B} + \mathbf{AB} = k \mathbf{I} = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

I used as or equated to $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ at some stage

B1

$(p+7=0, 14+2p=0 \Rightarrow) p = -7$

$p = -7$ provided it gives the relevant two zero elements

B1

$p = -7 \Rightarrow \mathbf{A} - \mathbf{B} + \mathbf{AB} = \begin{bmatrix} -27 & 0 \\ 0 & -27 \end{bmatrix} = -27 \mathbf{I}$

$\Rightarrow k = -27$

CSO

Either $-27 \mathbf{I}$ (no earlier errors) for B1

OR $k = -27$ with either $\begin{bmatrix} -27 & 0 \\ 0 & -27 \end{bmatrix}$ or $27\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ or $-27\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ seen before (no earlier errors) for B1

B1

4

[7]

Q5.

(a)

$$\begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

If A1 not scored, award M1A0 for all correct entries expressed in trig form eg

$$\begin{bmatrix} \cos 135 & -\sin 135 \\ \sin 135 & \cos 135 \end{bmatrix}$$

M1

A1

2

(b) (i)

$$\mathbf{M} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \sqrt{2} \times \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \left(\begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \right)$$

Or better

PI by cand. having both a correct scale factor of enlargement and a correct corresponding angle of rotation

M1

Scale factor of enlargement is $\sqrt{2}$

SF = $\sqrt{2}$ OE surd form

A1

Angle of rotation is 135 (degrees anticlockwise)

Angle = 135 OE eg -225

If M0 give B1 for SF = $\sqrt{2}$ OE surd and

B1 for angle = 135 OE

A1

3

(ii) For $\mathbf{M}^2$, SF of enlargement = 2

OE If incorrect, ft on [c's SF in (b)(i)]²

OE, eg -90(degrees), eg 90 (degrees) clockwise

B1F

Angle of rotation is 270 (degrees anticlockwise)

If incorrect, ft on $2 \times c$'s angle in (b)(i)

(neither B1F B1 nor B1 B1F is possible)

B1F

2

$$(iii) \quad M^2 = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

For complete method (matrix calculation or geometrical reasoning)

$$M^4 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix}$$

Matrix for M^2 could be seen earlier

(M0 if >1 independent error in matrix multiplication)

Geometrically SF = 4, rotation angle = 540

OE scores M1 and completion scores A1

M1

$$M^4 - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -4I$$

Either of these two forms convincingly shown

A1

2

$$(iv) \quad M^{2012} = (M^4)^{503} = (-4I)^{503} = -(2^2)^{503}I = -2^{1006}I$$

OE Fully explained, algebraically from $(-4I)^{503}$, or geometrically

E1

$$M^{2012} = -2^{1006}I$$

$$M^{2012} = -2^{1006}I (n = 1006) \quad (B0 \text{ if FIW})$$

B1

Geometrically: M^{2012} represents an enlargement with SF 2^{1006} followed by a rotation of angle $2012 \times 135^\circ$ ie 754.5 revolutions, being equivalent to rotation of 180° ie matrix is $-I$ so

$$M^{2012} = -2^{1006}I$$

2

[11]

Q6.

$$(a) \quad (i) \quad A^T = \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$$

B1

Good multiplication attempt at $\mathbf{A} \mathbf{A}^T$

If they multiply $\mathbf{A}^T \mathbf{A}$ instead, they can score B1 M1 A0 A0 A1 A1

A1A1

M1

$$= \begin{bmatrix} k^2 + 5 & 3k + 2 & 3k + 2 \\ 3k + 2 & k^2 + 5 & 3k + 2 \\ 3k + 2 & 3k + 2 & k^2 + 5 \end{bmatrix}$$

Main diagonal correct

All others correct

A1

A1

$$k = -\frac{2}{3} \quad m = 5\frac{4}{9} \text{ or } \frac{49}{9}$$

6

$$(ii) \quad \begin{bmatrix} -\frac{2}{3} & 1 & 2 \\ 2 & -\frac{2}{3} & 1 \\ 1 & 2 & -\frac{2}{3} \end{bmatrix}^{-1} = \frac{9}{49} \begin{bmatrix} -\frac{2}{3} & 2 & 1 \\ 1 & -\frac{2}{3} & 2 \\ 2 & 1 & -\frac{2}{3} \end{bmatrix}$$

ft their m and their k

B1

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$$\frac{9}{49} \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$$

Accept since the value of k is now known,

but not just $\frac{9}{49} \mathbf{A}^T$

Decimal version (correct to at least 3sf) is also ok:

$$\begin{bmatrix} -0.122 & 0.367 & 0.184 \\ 0.184 & -0.122 & 0.367 \\ 0.367 & 0.184 & -0.122 \end{bmatrix}$$

1

(b) (i) $\det \mathbf{A} = k^3 - 6k + 9$

Good attempt (cubic)

M1

A1

2

(ii) $\det \mathbf{A} = (k + 3)(k^2 - 3k + 3)$

Factorisation attempt

M1

$$k = -3$$

A1

$$\Delta = 9 - 12$$

B1

Special Cases

$k = -3$ but no working

(B1)

For all 3 roots, -3 and $\frac{3+i\sqrt{3}}{2}$ given
with no supporting working

(B3)

3

- (iii) Replacing k by $(k - 7)$
NOT just in the determinant form (ie starting again)

M1

$k = 4$ (however obtained)
ft their previous $k + 7$

A1

2

EXAM PAPERS PRACTICE [14]

Q7.

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$$\mathbf{X}^2 = \begin{bmatrix} 23 & 1 & -3 \\ 2 & 24 & 3 \\ -4 & 2 & 19 \end{bmatrix}$$

(a) (i)

≥ 5 correct for the M

M1

All 9 correct for the A

A1

$\mathbf{X}^2 - \mathbf{X} = 20\mathbf{I}$ i.e. $k = 20$
Shown legitimately

A1

3

- (ii) Mult⁹. $\mathbf{X}^2 - \mathbf{X} = 20\mathbf{I}$ by $\mathbf{X}^{-1}$

M1

Re-arranging $\mathbf{X} - \mathbf{I} = 20 \mathbf{X}^{-1}$

to get $\mathbf{X}^{-1} = \frac{1}{20}(\mathbf{X} - \mathbf{I})$

Legitimately

A1

2

(b) $\mathbf{X}^{-1} = \frac{1}{20} \begin{bmatrix} 2 & 1 & -3 \\ 2 & 3 & 3 \\ -4 & 2 & -2 \end{bmatrix}$

Noted or used

B1

$(\mathbf{XY})^{-1} = \mathbf{Y}^{-1} \mathbf{X}^{-1}$

Incl. attempt at the multn.

M1

$\begin{bmatrix} 6 & 3 & -9 \\ 2 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix}$

A1

3

[8]

Q8.

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(a) (i) $\mathbf{A}^2 = \begin{bmatrix} -2 & 2\sqrt{3} \\ -2\sqrt{3} & -2 \end{bmatrix}$

M1 if at least two entries correct

M1A1

2

(ii) $\mathbf{A}^3 = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$

if at least two entries correct

M1

..... = 81

A1

2

(b) (i) $\mathbf{A}^3$ gives enlargement with SF 8 (centre the origin)

M1 for enlargement (only);

ft wrong value for k

M1A1F

2

- (ii) Enlargement and rotation
Some detail needed

M1

Enlargement scale factor 2

A1

Rotation through 120° (antic'wise)

A1

3

[9]

Q9.

(a) $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

stated or used at any stage

B1

Attempt at $(A - I)^2$
with at most one numerical error

M1

$(A - I)^2 = \begin{bmatrix} 0 & 4 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ 3 & 0 \end{bmatrix} = 12I$

A1

3

(b) $(A - B) = \begin{bmatrix} 0 & 1 \\ 3 - p & 0 \end{bmatrix}$

B1

$(A - B)^2 = \begin{bmatrix} 3 - p & 0 \\ 0 & 3 - p \end{bmatrix}$

M1A0 if 3 entries correct

M1A1

... = $(A - I)^2$ for $p = -9$
ft wrong value of k

A1F

4

[7]

Q10.

(a)
$$\begin{bmatrix} 32 - 4t & 0 & 0 \\ 98 - 2t^2 & 4t - 24 & 2t - 14 \\ 28 - 4t & 0 & 4 \end{bmatrix}$$

*Decent attempt at **AB***

M1

≥ 5 correct

A1

all correct

A1

$t = 7$

*Allow this ft if only 1 or 2 elements of **AB** incorrect*

A1

$\mathbf{AB} = 4\mathbf{I}$

*Must be from **AB** completely correct*

A1

5

(b)
$$\mathbf{A}^{-1} = \frac{1}{4} \begin{bmatrix} 115 & -4 & -1 \\ -14 & 4 & 2 \\ 17 & -4 & -3 \end{bmatrix}$$

*ft 1/det if related to **B***

B1

CAO; must be "deduced"

$\frac{1}{4}\mathbf{B}$
NB " $\frac{1}{4}\mathbf{B}$ " scores B1 only

B1

2

[7]

Q11.

(a) Rotation 45° (anticlockwise) (about O)

M1 for 'rotation'

M1A1

2

(b) Reflection in $y = x \tan 22.5^\circ$

M1 for 'reflection'

M1A1

2

- (c) Rotation 90° (anticlockwise) (about O)
*M1 for 'rotation' or correct matrix;
 ft wrong angle in (a)*

M1A1F

2

- (d) Identity transformation
ft wrong mirror line in (b); B1 for $\mathbf{B}^2 = \mathbf{I}$

B2,1F

2

- (e) $\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
allow M1 if two entries correct

M1A1

Reflection in $y = x$

A1

3

[11]

EXAM PAPERS PRACTICE

Q12.

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- (a) (i) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 0 & 2k \\ 2k & 0 \end{bmatrix}$

B1

1

- (ii) $\mathbf{A}^2 = \begin{bmatrix} 2k^2 & 0 \\ 0 & 2k^2 \end{bmatrix}$
B1 if three entries correct

B2,1

2

- (b) $(\mathbf{A} + \mathbf{B})^2 = \begin{bmatrix} 4k^2 & 0 \\ 0 & 4k^2 \end{bmatrix}$
B1 if three entries correct

B2,1

$\mathbf{B}^2 = \mathbf{A}^2$, hence result

B1B1 4

- (c) (i) A^2 is an enlargement (centre O)

M1

with SF 2

Condone $2k^2$

A1 2

- (ii) Scale factor is now $\sqrt{2}$
Condone $2k$

B1

Mirror line is $y = x \tan 22\frac{1}{2}^\circ$

M1 A1 3

[12]

Q13.

(a) (i) $M^2 = \begin{bmatrix} 12 & 0 \\ 0 & 12 \end{bmatrix}$

M1 if zeroes appear in the right places

M1A1

$= 12I$
ft provided of right form

A1F 3

(ii) $q \cos 60^\circ = \frac{1}{2}q = \sqrt{3} \Rightarrow q = 2\sqrt{3}$
OE SC $q = 2\sqrt{3}$ NMS 1/3

M1A1

Other entries verified
surd for $\sin 60^\circ$ needed

E1 3

(b) (i) $SF = q = 2\sqrt{3}$
ft wrong value for q

B1F 1

(ii) Equation is $y = x \tan 30^\circ$

B1

1

(c) $M^4 = 144I$

PI; ft wrong value in (a)(i)

B1F

M^4 gives enlargement SF 144

ft if c's $M^4 = kI$

B1F

2

[10]

Q14.

(a) (i) $M^2 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$

M1

$$= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix}$$

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$$\begin{aligned}
 M^2 + 2I &= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\
 &= \begin{bmatrix} 6 & -3 & 3 \\ 3 & 0 & 3 \\ 3 & -3 & 6 \end{bmatrix} = 3M
 \end{aligned}$$

ie $k = 3$

A1

3

(ii) Multiplying by M^{-1}

M1

to get $M + 2M^{-1} = 3I$

ft

A1

so that $\mathbf{M}^{-1} = \frac{3}{2}\mathbf{I} - \frac{1}{2}\mathbf{M}$
 ie $a = -\frac{1}{2}$ and $b = \frac{3}{2}$

A1

3

- (b) (i) Char. eqn. is $\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$
One A mark for each of the other coefficients

M1A1A1A1

ie $(\lambda - 2)(\lambda - 1)^2 = 0$
Good factorisation attempt

M1

giving $\lambda_1 = 1$ (twice) and $\lambda_2 = 2$

A1

6

(ii) $\lambda = 1 \Rightarrow x - y + z = 0$ (thrice)

B1

Any two independent eigenvectors
Attempted

M1

(eg) $\alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

A1

$\lambda = 2 \Rightarrow -y + z = 0$
 $x - 2y + z = 0 \Rightarrow x = y = z$
 $x - y = 0$

M1

$y \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

A1

5

- (iii) For $\lambda = 1$, eigenvectors represent a plane of invariant points
Plane

M1A1

For $\lambda = 2$, eigenvectors represent an invariant line

B1

3

[20]

Q15.

(a) $\mathbf{AB} = \begin{bmatrix} 0 & -4 \\ 4 & 0 \end{bmatrix}$

M1A0 if 3 entries correct

M1A1

2

(b) $\mathbf{A}^2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

$\dots = 4\mathbf{I}$

B1

B1

2

(c) $(\mathbf{AB})^2 = -16\mathbf{I}$

$\mathbf{B}^2 = 4\mathbf{I}$

PI

B1

B1

so $\mathbf{A}^2 \mathbf{B}^2 = 16\mathbf{I}$ (hence result)

Condone absence of conclusion

B1

3

[7]

Q16.

(a) (i) $\det \mathbf{P} = 4a + 6 + 4 + a = 5a + 10$

M1A1

2

(ii) When $a = 3$, $\det \mathbf{P} = 25$

ft

B1F

1

(iii) Setting their $\det \mathbf{P} = 0 \Rightarrow a = -2$

M1

ft

A1F

2

(b) (i) $\mathbf{P}^{-1} = \frac{1}{25} \mathbf{Q}$

B1

1

(ii) $(\mathbf{PQ})^{-1} = (25 \mathbf{I})^{-1} = \frac{1}{25} \mathbf{I}$

M1A1

2

Or $(\mathbf{PQ})^{-1} = \mathbf{Q}^{-1} \mathbf{P}^{-1}$

Ignore $(\mathbf{PQ})^{-1} = \mathbf{P}^{-1} \mathbf{Q}^{-1}$ if they can make it work

(M1)

$= \mathbf{Q}^{-1} \cdot \frac{1}{25} \mathbf{Q} = \frac{1}{25} \mathbf{I}$

(A1)

(2)

(iii) $\det \mathbf{PQ} = \det (25 \mathbf{I}) = 25^3$ or 15625

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M1A1

$\det \mathbf{PQ} = \det \mathbf{P} \cdot \det \mathbf{Q}$

Used

M1

$\Rightarrow 25^3 = 25 \det \mathbf{Q}$

$\Rightarrow \det \mathbf{Q} = 25^2$ or 625

A1

4

[12]

Q17.

(a) $\mathbf{M} = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$

B1 if subtracted the wrong way round

B2,1

2

(b) $p = 3$

ft after B1 in (a)

B1F

L is $y = -x$

Allow $p = -3$, L is $y = x$

B1

2

(c) $\mathbf{M}^2 = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$

Or by geometrical reasoning; ft as before

B1F

$\dots = 9\mathbf{I}$

ft as before

B1F

2

[6]

Q18.

(a) (i) Reflection ...

M1

$\dots$ in $y = -x$

OE

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A1

2

(ii) $\mathbf{A}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

M1A0 for three correct entries

M1A1

2

(iii) $\mathbf{A}^2 = \mathbf{I}$ or geometrical reasoning

E1

1

(b) (i) $\mathbf{B}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

M1A0 for three correct entries

M1A1

$$\mathbf{b}^2 - \mathbf{A}^2 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

ft errors, dependent on both M marks

A1ft

3

$$(ii) \quad (\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A}) = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

B1

$$\dots = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

ft one error; M1A0 for three correct (ft) entries

M1

A1✓

3

[11]

Q19.

$$(a) \quad (i) \quad \mathbf{M}^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

M1 if 2 entries correct

M1

M1A1 if 3 entries correct

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A2,1

3

$$(ii) \quad \mathbf{M}^4 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

ft error in $\mathbf{M}^2$ provided no surds in $\mathbf{M}^2$

B1ft

1

(b) Rotation (about the origin)

M1

... through 45° clockwise

A1

2

(c) Awareness of $\mathbf{M}^8 = \mathbf{I}$

OE; NMS 2/3

M1

$$M^{2006} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

complete valid method

m1

ft error in M^2 as above

A1ft

3

[9]

Q20.

- (a) Setting $\det \mathbf{P} (2t - 6 + 2 - 3t + 8 - 1 = 3 - t) = 0$

M1

$$\Rightarrow t = 3$$

A1

2

(b) (i)
$$\begin{bmatrix} 6 & 0 & 0 \\ -7t - 21 & 3 - t & 15 + 5t \\ 0 & 0 & 6 \end{bmatrix}$$

Multiplication attempt with ≥ 3 correct elements

M1

Any one row or column

A1

All correct

A1

3

- (ii) When $t = -3$, $\mathbf{PQ} = 6\mathbf{I}$

The 6 may be implicit

B1 B1

2

(iii)
$$\mathbf{Q}^{-1} = \frac{1}{6} \mathbf{P} = \frac{1}{6} \begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & -2 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\frac{1}{6} \mathbf{P} \text{ or } \frac{1}{36} (6\mathbf{P})$$
Allow for

B1

1

- (c) Enlargement (centre O) sf 6

M1 A1

2

[10]

Q21.

(a) (i) $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 3 & 6 \\ 6 & 3 \end{bmatrix}$

M1A0 if 3 entries correct

M1A1

2

(ii) $\mathbf{AB} = \begin{bmatrix} 8 & 6 \\ 6 & 8 \end{bmatrix}$
Ditto

M1A1

2

(b) $\mathbf{A} + \mathbf{B} - \mathbf{AB} = \begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}$

ft wrong answers in (a)

B1F

$\dots = -5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B1

2

[6]

Examiner reports

Q1.

Only half of the students were able to make satisfactory progress with this question. Most

of the successful attempts substituted $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (or equivalent) for **A**, and then solved the resulting simultaneous equations. Only the most able students were able to successfully isolate **A** or **A**⁻¹ by rearranging the given matrix equation.

Q2.

In this question on matrices, parts (a)(i) and (a)(ii) were almost always answered correctly,

although some students left their answer to (a)(ii) in the form $\begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$ without indicating that they clearly appreciated what **I** represented. A correct description of the geometrical transformation represented by **A** was usually given although reflection in the line $y = x$ was the most frequent incorrect answer.

Most students scored the method mark in part (b)(ii) but some then gave the incorrect value $\sqrt{2}$ for k . Part (b)(iii) proved to be more of a challenge. A significant minority of students could not make any real progress. Most realised that the matrices **A** and **B**

needed to be multiplied but the product was not always linked with the matrix for **P**, $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$, in a correct manner. Some better approaches just made the mistake of using **BA** instead of **AB**. For the final mark the examiners wanted the coordinates of the image of **P** to be written in coordinate form rather than left as a matrix.

Q3.

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This question elicited a good response from students. There was evidence of efficient use of graphic calculators in parts (a) and (c). In part (b), students had to produce a watertight proof to get both marks. It was disconcerting to find students writing “divide by **A**” or “cancel **A**”, displaying a lack of understanding matrix algebra. In the final part, many chose not to use the inverse matrix to find the solutions and instead chose to use elimination. Varying degrees of success were seen – the most successful chose to eliminate y from the second and third equations.

Q4.

Many correct answers were submitted for part (a) in this question which tested 2×2 matrices including the identity matrix **I**. In part (b) the most successful starting approach was to find the matrix for **A** – **B** + **AB** in terms of p , equate it to the matrix for k **I** to show that $p = -7$ and then to work with **A** – **B** + **AB** and show that it was equal to -27 **I**. A minority of students found the correct value of k but either did not convincingly show that **A** – **B** + **AB** = k **I** or made an incorrect concluding statement. It was disappointing to see some students writing down an incorrect matrix for **I**.

Q5.

This was the least well-answered question on the paper with only the most able scoring full marks. Part (a) was generally well-answered with only a small minority failing to give

the matrix in surd form. In part (b)(i), better students showed clear working which related to part (a), and stated correct answers for both the scale factor of the enlargement and the angle of the rotation.

In part (b)(ii) the most common method seen was to find the matrix M^2 first and use it to find the scale factor and angle rather than stating the scale factor to be the square of the scale factor in part (b)(i) and the angle to be double the angle in part (b)(i). Those using the matrix M^2 usually gave the correct scale factor but 90° anticlockwise was a common wrong angle.

Students generally scored both marks in part (b)(iii), although some others made an arithmetic error in their matrix multiplication or did not complete the question appropriately and thus lost the final mark. Part (b)(iv) proved to be very challenging and only a small number showed the deduction sufficiently rigorously. There were many examples of students obtaining the correct value of n but from incorrect reasoning. Most attempted this part using algebraic methods but relatively few gave due consideration to the negative sign.

Some excellent deductions based on geometric reasoning using the earlier results in the question were also seen.

Q6.

Apart from the few who did not seem to know what a transpose matrix was, this was a popular and high-scoring question, even for those with relatively low scores on the paper as a whole. The only really disturbing aspect of the work on display was that in part (b)(ii) so few students seemed to have any idea how to go about finding a linear factor of a fairly simple cubic and then factorising to find the accompanying quadratic factor, which is C1 work. Most students who gained the answer $k = -3$ appeared to get it from a calculator, and very few indeed could use it to find the required quadratic factor; fewer still were able to reason convincingly that this quadratic factor had no real roots, which again is C1 work.

In the final part of the question, many students failed to realise that k was now being replaced in the given matrix A by $k - 7$, so that all they had to do was to add 7 to their previous value of k for two straightforward marks. Almost all students who correctly found the answer for this new k did so by going back, working out a cubic characteristic equation (with some large numbers for coefficients), and using a calculator to solve it for them. This was, somewhat reluctantly, awarded the marks if the answer was correct. Most of those students who did not get the correct answer also used this method, but any incorrect working prevented their calculator helping them out and so they gained no marks.

Q7.

This was another popular question, in that responses were generally at least partially successful. Even so, around half of all candidates failed to realise that the “hence” in part (a)(ii) meant that they were supposed to work algebraically with the result of part (a)(i) rather than go through the lengthy approach of finding X^{-1} directly. About the same proportion incorrectly opted for $(XY)^{-1} = X^{-1}Y^{-1}$. In addition to all these errors, a lot of marks were lost due to carelessness; mistakes made in calculating X^2 were often not corrected, even when the candidates clearly failed to arrive at a multiple of I in part (a)(i).

Then, despite being told that X^{-1} could be found by $\frac{1}{20}(X - I)$ in part (b), many made a slip

somewhere in the working, and the correct $(XY)^{-1}$ appeared far less often than should

have been the case, given the information given in the question.

Q8.

The first matrix calculation in this question involved a fair number of surds and minus signs, and yet it was very accurately performed by the majority of candidates. After that, it was relatively easy to reach a matrix of the required form in part (a)(ii). Almost all candidates knew how to describe the transformation represented by this matrix, but by contrast many had a long and unsuccessful struggle to describe the transformation required in part (b)(ii). An acceptable alternative here was an enlargement with scale factor -2 and a rotation through 60° clockwise.

Q9.

As usual on this paper, the work on matrices was very good indeed, with most candidates working out all the steps efficiently. Some tried to expand the expressions $(\mathbf{A} - \mathbf{I})^2$ and $(\mathbf{A} - \mathbf{B})^2$ but almost invariably assumed commutativity of multiplication. A rather silly way to lose a mark was to work correctly to the equation $3 - p = 12$ and then to solve this equation incorrectly, which happened quite frequently.

Q10.

The matrix multiplication was generally handled very well, apart from a small but noticeable number of candidates who insisted on attempting $\mathbf{BA}$ despite the question. The only other obstacle to a completely correct part (a) was found amongst those candidates who had the odd one or two incorrect elements, which they had failed to notice and correct due to a lack of a check, meaning that $t = 7$ did not actually give $4\mathbf{I}$ consistently for their $\mathbf{AB}$. As with question 2, the word “deduce” in part (b) was almost totally ignored, and alternative methods for finding an inverse, often taking up lots of time, received no credit.

Q11.

High marks were often earned in this question, generally from the multiplication of the matrices rather than from the geometrical explanations, which tended to be shaky. In parts (c) and (d) the vast majority of candidates calculated a matrix product rather than base their answers purely on the transformations already found in parts (a) and (b). The transformation in part (c) was often given as a reflection rather than a rotation, and in part (d) many candidates stated that the matrix was the identity matrix but did not make any geometrical statement as to what this matrix represented. In part (e) the correct matrix was often obtained but the candidates failed to give the correct geometrical interpretation, or in some cases resorted to a full description of the transformation as a combination of the reflection and rotation found in parts (b) and (a). When this was done correctly, full credit was given.

Q12.

The first nine marks in this question seemed to be found very easy for most candidates to earn. The presence of so many k 's in the matrices did not prevent them from carrying out all the calculations correctly and confidently, though sometimes the flow of the reasoning in part (b) was not made totally clear. Part (c)(ii), on the other hand, proved too hard for many candidates, who seemed to resort to guesswork for their answers.

Q13.

Part (a)(i) of this question was almost universally well answered, but part (a)(ii) led to

some very slipshod and unclear reasoning. In part (b)(i) many candidates did not appreciate that the scale factor of the enlargement must be the same as the value of q obtained in the previous part. In part (b)(ii) a common mistake was to use $\tan 60^\circ$ in finding the gradient of the mirror line instead of halving the angle to obtain $\tan 30^\circ$. Answers to part (c) were mostly very good, even from candidates who had been struggling with the earlier parts of the question.

Q14.

This question was definitely the most demanding one on the paper; partly for its length, and partly due to working with a 3×3 matrix. On top of this, there was a repeated eigenvalue to the matrix, and a lot of candidates clearly did not know quite what to make of it. Nonetheless, even the relatively weaker candidates were still able to gain up to 12 of the marks.

Part (a)(i) was very well handled by those who attempted it. Part (a)(ii) was less well handled. Many of those who fell down here did so because they then went off to find the inverse matrix $\mathbf{M}^{-1}$ itself, rather than follow the logic of the question. Many more who did multiply the matrix equation by $\mathbf{M}^{-1}$ got tangled up in some way – failing to spot that $\mathbf{M}\mathbf{M}^{-1} = \mathbf{I}$, for instance, or being unable to divide by 2 under exam conditions.

In part (b), the factorisation of the determinant $|\mathbf{M} - \lambda\mathbf{I}|$ was usually approached by row/column operations, which was very pleasing to see. However, it frequently (actually usually) led to the strange sight of candidates taking a partially or completely factorised expression, and then multiplying it out to get a standard cubic form, only to start to factorise it again to find the three eigenvalues.

Those who made a mistake in this working then shot themselves in the foot, losing the following method mark for the attempt to factorise, which they failed to pick up by presuming that their cubic actually was equal to the factorised form $(\lambda - 1)^2(\lambda - 2)$ rather than doing any of the necessary working or going back to check their previous, incorrect working.

The final two parts to the question were demanding, but actually not as difficult as candidates found them to be. Despite the fact that most candidates getting this far quite correctly deduced that $x - y + z = 0$ for the repeated eigenvalue $\lambda = 1$, almost all of them then believed this represented the equation of a line. This error then prevented them from finding a pair of independent representative eigenvectors (or an alternative parametric plane form) and also from describing the result as a plane of invariant points, rather than as a line.

Q15.

This question was very well answered by the majority of candidates, who showed confidence and accuracy in manipulating these simple matrices.

Q16.

This was the last question on the paper because, since it involves a 3×3 matrix, it was potentially the most time-consuming question. However, candidates found the structure of the question very helpful.

Q17.

This question was generally well answered. The most common errors were in part (b), where the mirror line was given as $y = x$ rather than $y = -x$. It was, however, acceptable to

give this reflection in conjunction with an enlargement with scale factor -3 .

In part (c) some candidates omitted the factor 3 before carrying out the matrix multiplication, while others multiplied incorrectly to obtain the 9s and 0s in all the wrong places.

Q18.

Many candidates were successful in part (a)(i) of this question. Wrong answers usually involved a 180° rotation or a reflection in $y = x$. Part (a)(ii) was nearly always done correctly, but the explanations offered in part (a)(iii) were often wrong or too vague to be worth a mark. The matrix calculations in part (b) were done well apart from occasional careless slips. Some candidates seemed to think that three marks were being offered in part (b)(ii) for merely quoting the same answer as in part (b)(i), while others attempted to use matrix algebra to express $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A})$ in an alternative form before evaluating their expression. This usually led to a wrong answer.

Q19.

Most candidates carried out the matrix multiplications accurately in part (a) of this question. In part (b) many candidates stated correctly that the required transformation was a rotation, but often gave the angle as 45° anticlockwise rather than clockwise. Attempts at part (c) were rarely completely successful. Many candidates found that $\mathbf{M}^8$ was the identity matrix but then went on to say that $\mathbf{M}^{2006}$ must also be the identity matrix. Others gave the same answer as in part (a) (i), or gave the correct answer without any indication as to their reason for doing so, or with incomplete reasoning.

Q20.

This proved to be a profitable question for many candidates. In particular, the multiplication of two 3×3 matrices was handled very competently indeed, even though algebraic terms were involved. The odd error arising here did not prevent candidates from doing good work in the later parts of the question. They were not required to check that all entries of the product matrix gave consistent values for both t and k .

Q21.

Most candidates made a good start to the paper by answering this question confidently, although in part (c) many candidates failed to show knowledge of the identity matrix.